Review

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TimeBook®

Common Units and Their Rates

Decimals, Fractions, Percentages

Rectangular Prisms and Cubes

Circles

Venn Diagram

Ratio

Cylinders and Cones

Permutation and Combination

Pigeonhole Principle

Common Units and Their Rates

Common Units and Their Rates

1. **Length Units:**

kilometer (km) $\xrightarrow{1000}$ meter (m) $\xrightarrow{10}$ decimeter (dm) $\xrightarrow{10}$ centimeter (cm) $\xrightarrow{10}$ millimeter (mm)

2. **Area Units:**

Square kilometer (km^2) $\xrightarrow{100}$ hectare (ha) $\xrightarrow{10000}$ Square meter (m^2) $\xrightarrow{100}$ Square decimeter (dm^2) $\xrightarrow{100}$ Square centimeter (cm^2)

3. **Volume Units:**

Cubic meter (m^3) $\xrightarrow{1000}$ Cubic decimeter (dm^3) $\xrightarrow{1000}$ Cubic centimeter (cm^3)

4. **Capacity Units:**

liter(l) $\xrightarrow{1000}$ milliliter(ml)

Cubic decimeter (dm^3) = liter(l); Cubic centimeter (cm^3) = milliliter(ml)

5. Time Units:

Common year 365 days

Leap year 366 days

Century $\xrightarrow{100}$ year $\xrightarrow{12}$ month $\rightarrow$ day $\xrightarrow{24}$ hour $\xrightarrow{60}$ minute $\xrightarrow{60}$ second

- Long Month (31d): 1、3、5、7、8、10、12
- Short Month (30d): 4、6、9、11
- Common year Feb (28d)
- Leap year Feb (29d)

6. Mass Units: Ton (t) $\xrightarrow{1000}$ Kilogram (kg) $\xrightarrow{1000}$ gram (g)

7. Currency Units: dollar $\xrightarrow{10}$ dime $\xrightarrow{10}$ cent

Fill in the blanks

0.4m=(40)cm	$0.24km^2$=(240000)m^2
725mm=(7.25)dm	$4160cm^2$=(41.6)dm^2
$3.2m^2$=(320)dm^2	2.8L=(2800)mL
$5dm^2$=(500)cm^2	$8.75m^3$=(8750)dm^3
7500mL=(7.5)L	$64cm^3$=(0.064)dm^3
$6.2dm^2$=(620)cm^2	320mL=(320)cm^3

Decimals, Fractions, Percentages

- **What happens to the size of a decimal when the decimal point is moved to the right ?**

Move the decimal point two places to the right.

0.01 dollar $\xrightarrow{\text{One place to the right}}$ 0.10 dollar $\xrightarrow{\text{One place to the right}}$ 1.00 dollar

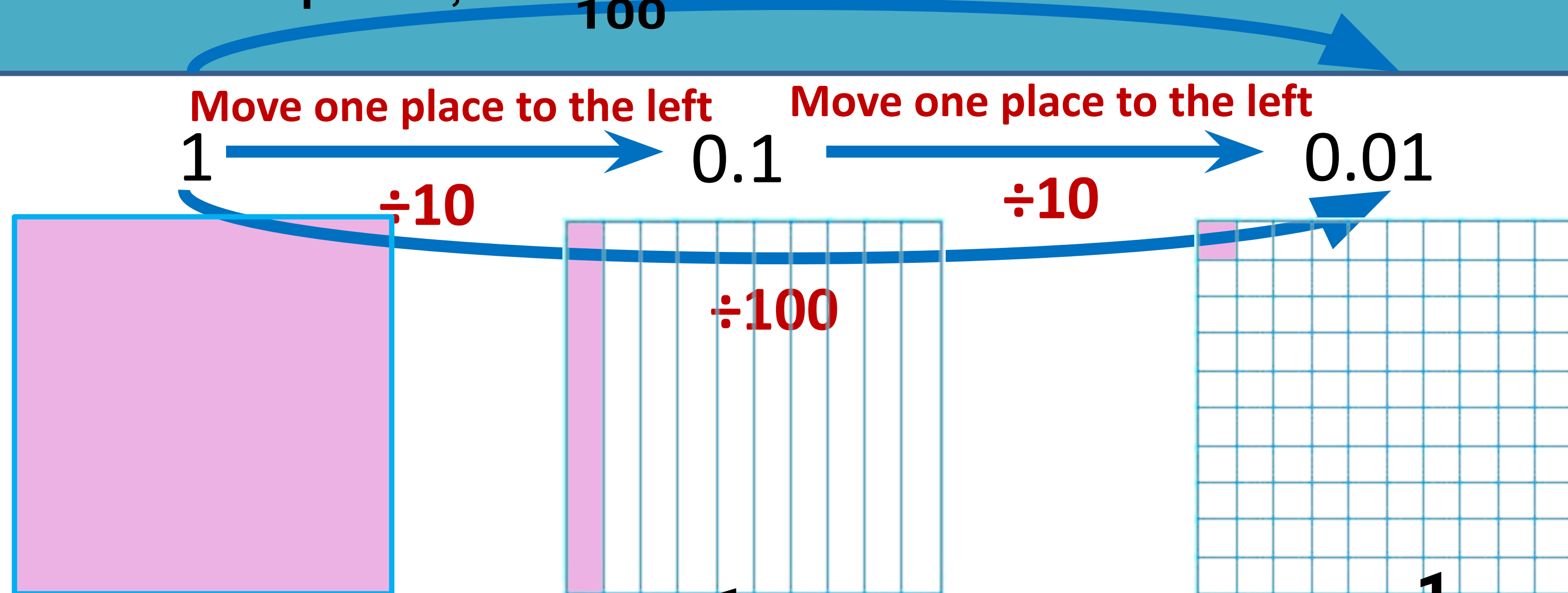
1 cent $\xrightarrow{\times 10}$ 1 dime $\xrightarrow{\times 10}$ 1 dollar

×100

What happens to the size of a decimal when the decimal point is moved to the left ?

When the decimal point of a number is moved one place to the left, the resulting number is $\frac{1}{10}$ of the original number

When moved two places, it is $\frac{1}{100}$



0.1 is $\frac{1}{10}$ of 1

0.01 is $\frac{1}{100}$ of 1

When the decimal point of a number is moved one place to the right, the resulting number is 10 times the original number .

When the decimal point of a number is moved two places to the right, the resulting number is 100 times the original number .

When the decimal point of a number is moved one place to the left, the resulting number is $\frac{1}{10}$ the original number .

When the decimal point of a number is moved two places to the left, the resulting number is $\frac{1}{100}$ the original number .

Practice

To determine the number of decimal places in the product, simply count the total number of decimal places in the two multipliers.

Place the decimal point in the following products.

$4.7 \times 1.5 = 705$

$9.32 \times 1.4 = 13048$

$1.87 \times 2.3 = 4301$

$1.6 \times 32 = 512$

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Explore

Wrapping a gift box requires 0.8 meters of wrapping paper and 2.4 meters of ribbon.

Wrapping paper: 2.6 dollars/ m
Ribbon: 0.85 dollars/m

How much does it cost to buy the ribbon?

$2.4\times0.85=\boxed{2.04}$ (**dollars**)

$$\begin{array}{r} 2.4 \\ \times 0.85 \\ \hline 120 \\ 192 \\ \hline 2040 \end{array}$$

One decimal place

Two decimal places

Three decimal places

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Practice

Place the decimal points in the products of the following problems.

$2.4\times2.8=672$

$4.23\times0.8=3384$

$40.2\times11=4422$

$0.15\times1.7=0255$

$0.4\times38.2=1528$

$0.56\times0.67=03752$

The calculation method for decimal multiplication

Ignore the Decimal Point: First, multiply the numbers as if they were whole numbers, ignoring the decimal points.

Multiply as Usual: Perform the multiplication just as you would with whole numbers.

Count the Decimal Places: After multiplying, count the total number of decimal places in the numbers you multiplied (i.e., the number of digits to the right of the decimal points in both numbers).

Place the Decimal Point: In the product, place the decimal point so that the number of decimal places in the result equals the total number of decimal places counted in step 3. If there are not enough digits, add zeros to the left of the product. If the product has trailing zeros, place the decimal point first and then remove the trailing zeros.

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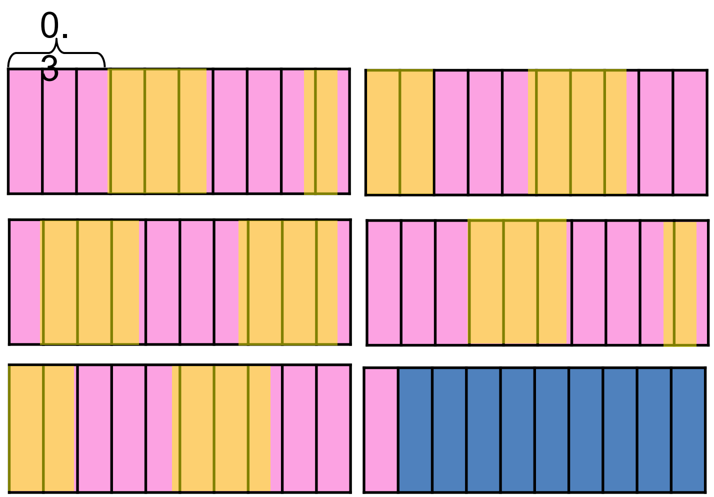
Explore

Domestic calls cost 0.3 dollars per minute, and the total cost was 5.1 dollars.

How many minutes of calls were there in total?



$5.1 \div 0.3 = 17$

$$\begin{array}{r} 17 \\ 0.3\overline{)5.1} \\ 3 \\ \hline 21 \\ 21 \\ \hline 0 \end{array}$$

Practice

$0.544 \div 0.16 = 3.4$　　$25.6 \div 0.032 = 800$

$$
\begin{array}{r}
3.4 \\
0.16 \overline{)\,0.54.4} \\
48 \\
\hline
64 \\
64 \\
\hline
0
\end{array}
\qquad
\begin{array}{r}
800 \\
0.032 \overline{)\,25.6\color{red}{00}} \\
256 \\
\hline
0
\end{array}
$$

To convert the divisor 0.032 into an integer, how many places should the decimal point be moved to the right?

Conclusion

When calculating division where the divisor is a decimal, based on the rule that 'the quotient remains unchanged when both the dividend and divisor are multiplied by the same factor (except 0),'

First remove the decimal point from the divisor to make it an integer.

Then, check how many decimal places the original divisor had, and move the decimal point in the dividend to the right by the same number of places. If there are not enough digits, add zeros to make up the difference.

Finally, perform the calculation according to the method for dividing decimals by an integer divisor.